

Welcome to the

Java Course

Module 1 – Day 1

Contents of the module

- **Introduction to programming**
- **Basic programming concepts**
- **Variables and Data Types**
- Conditionals
- Loops
- Control Structures
- Introduction to algorithms

About Me

Fuad Mahmudov

Professional Background

- 5+ years as a Java developer
- Built enterprise and web applications
- Experience in teaching and mentoring beginners



<https://www.linkedin.com/in/fuad-mahmudov/>

What about you ?

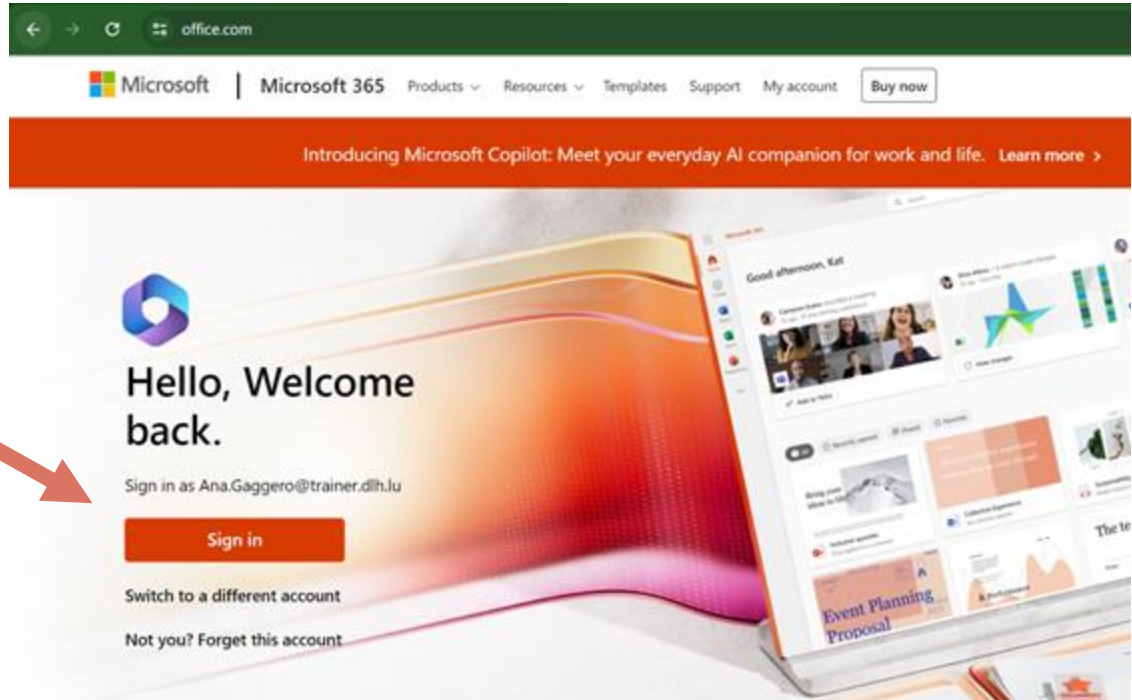
1. Name
2. Background
3. What brings you to Java Module 1?
4. What do you expect from this course?

Log into the local PC and to Microsoft Teams

Open Google Chrome and type: office.com

Step 1: Click on “Sign in”

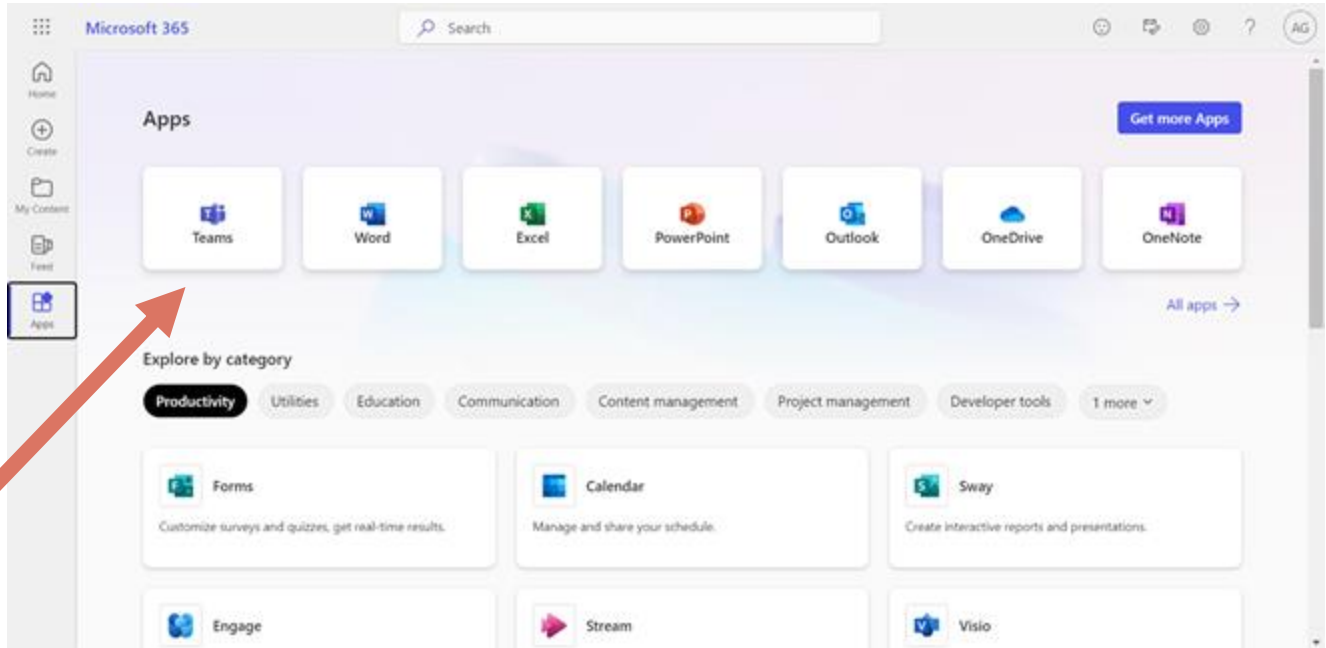
You should have received your password by email.



Log into the local PC and to Microsoft Teams

Step 2: Click on “Teams”

1



2

Log into the virtual machine

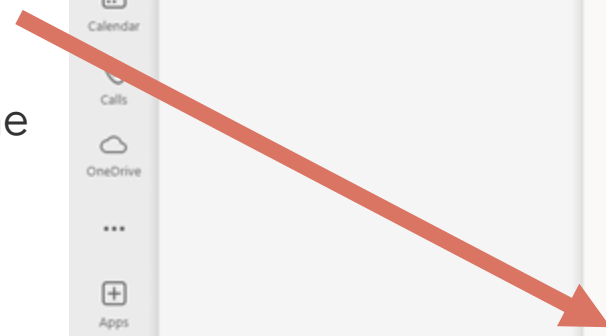
Step 1



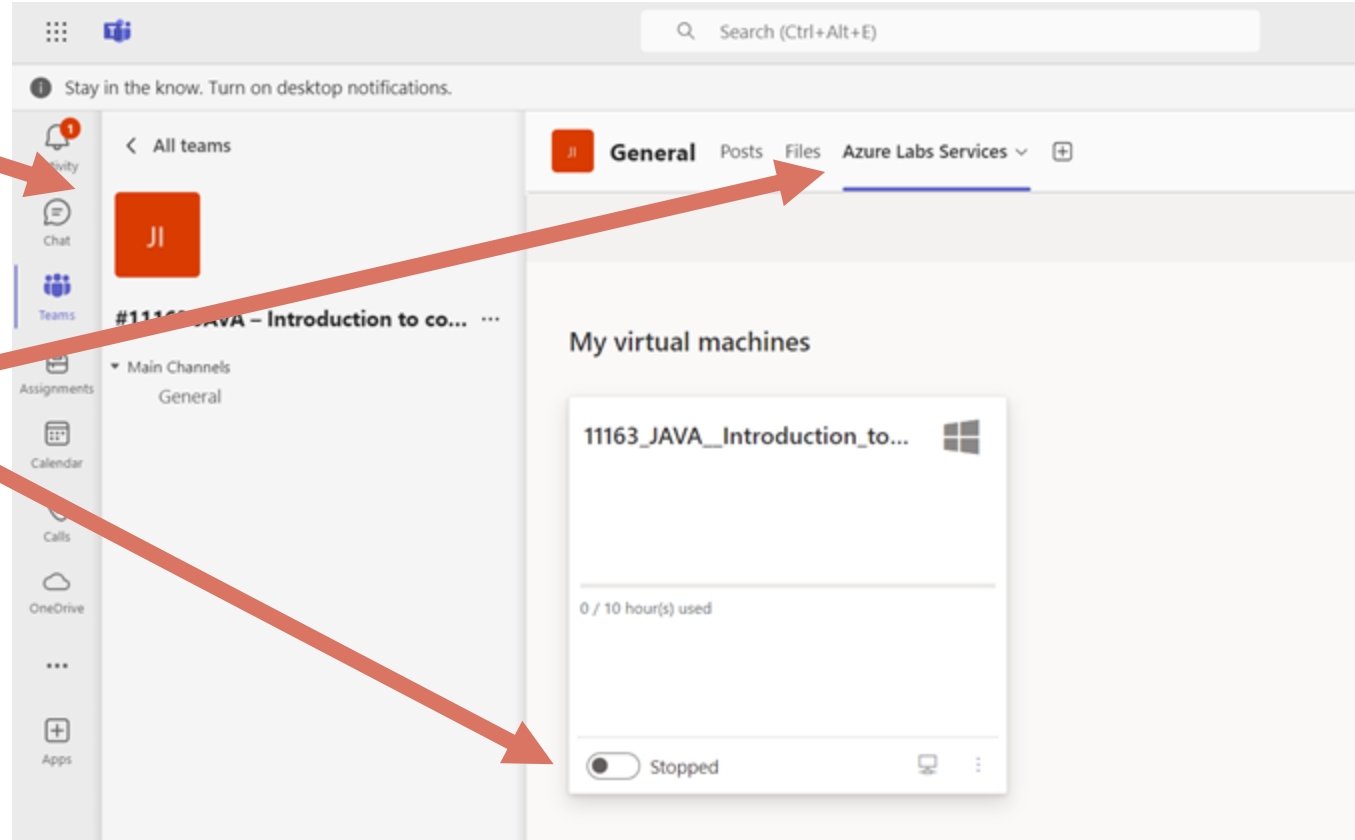
Step 2



Step 3



The virtual machine will be already on. If not, turn it on, it can take up to 5 minutes to start.



Log into the virtual machine

Username: student

Password: StudentDLH!

Project Students

Building a program that **gathers information** from students and **validates** it.

1. **Student Registration**
2. **Data Validation**

Project Tic Tac Toe

A program that
allows two players
to play Tic Tac Toe
against each other

```
What is your name Player 1? Ana  
What is your name Player 2? Juan
```

```
Ana will be X and Juan will be O
```

```
1 | 2 | 3  
4 | 5 | 6  
7 | 8 | 9
```

```
Ana choose your next move: 4
```

```
1 | 2 | 3  
X | 5 | 6  
7 | 8 | 9
```

Computational Thinking

Problem-solving approach
that helps you to think
logically and solve problems
effectively.

Computational Thinking

Decomposition:

Breaking down a complex problem into smaller, more manageable sub-problems or tasks.

Computational Thinking

Pattern Recognition:

Identifying recurring patterns or similarities in different problems to develop generic solutions.

Computational Thinking

Abstraction:

Focusing on essential details while ignoring irrelevant information to simplify the problem.

Computational Thinking

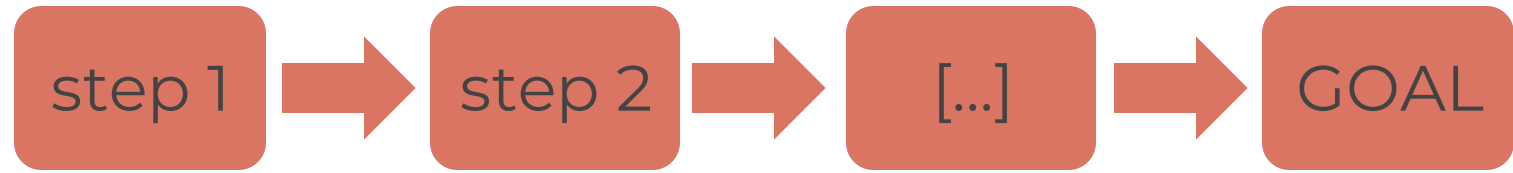
Evaluation and Optimization:

Evaluate the efficiency and effectiveness of solutions and refining them for better performance.

Algorithm

An Algorithm is an ordered **series of steps** that leads to the achievement of a goal or the solution to a problem.

Algorithm



Computational Thinking

It is not limited to computer science but can be applied to various disciplines and **real-world challenges.**

Computational Thinking

Make your sandwich

Decompose

step by step



Computational Thinking

1. **Prepare the ingredients**
2. **Assemble the ingredients**
3. **Plate it**



Concept: Decomposition and Abstraction

Computational Thinking

1. Prepare the ingredients

1. **Cut the slices of bread**

2. **Cut tomato**

3. **Take the ham out of the fridge**

4. **Take the salad out of the fridge**



Computational Thinking

1. Prepare the ingredients
2. Assemble the ingredients

1. **Slice of bread**

2. **Salad**

3. **Ham**

4. **Tomato slices**

5. **Slice of bread**



Computational Thinking

1. Prepare the ingredients
2. Assemble the ingredients
3. Plate it

1. Put it on a plate



Concept: Algorithm

Computational Thinking

Now, make a Lasagna

1. Prepare the ingredients
2. Assemble the ingredients
3. ~~Plate it~~ Cook it in the oven



Concept: Pattern

Programming Language

is a language that humans can use and that the machine can understand.

Programming Language



Swift



Java

One of the **most used** programming language. For example, it's used in the development of **desktop** and **mobile** applications.



MINECRAFT



Spotify®



Code Editor

To develop in a programming language, you need a code editor that understands and executes the language. In this course we will mainly use a very common one:

Visual Studio Code



Output

```
System.out.println("text");
```

Let's create your first

Hello world !

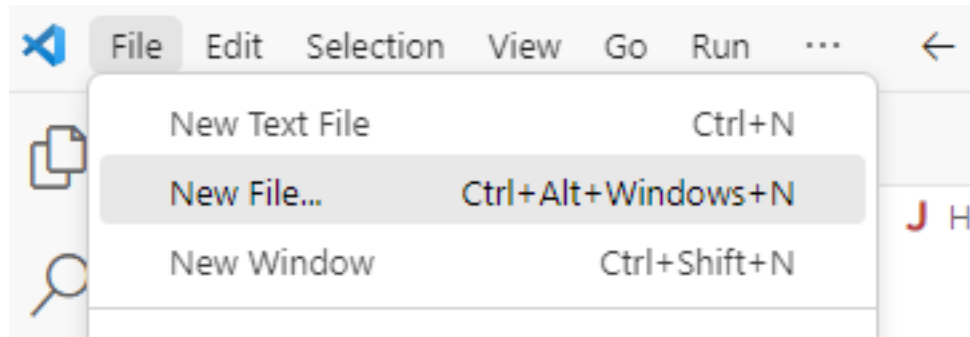
Code Editor

Create a new program



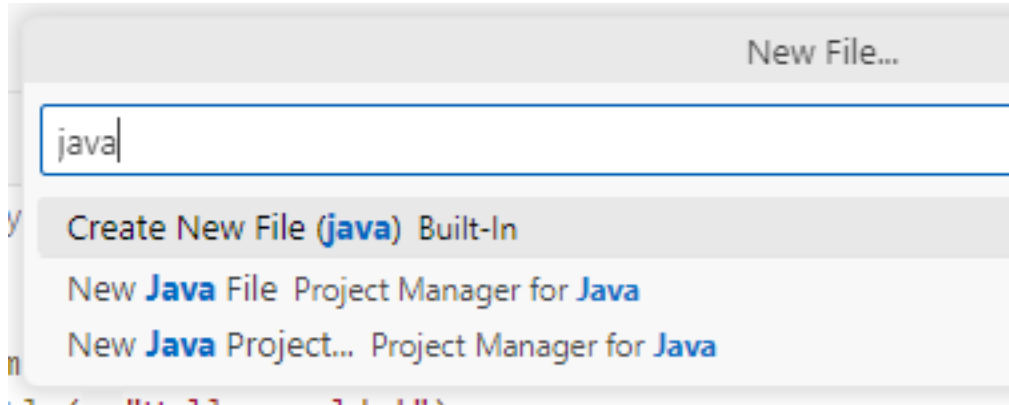
Code Editor

Step 1: **Create** a file.



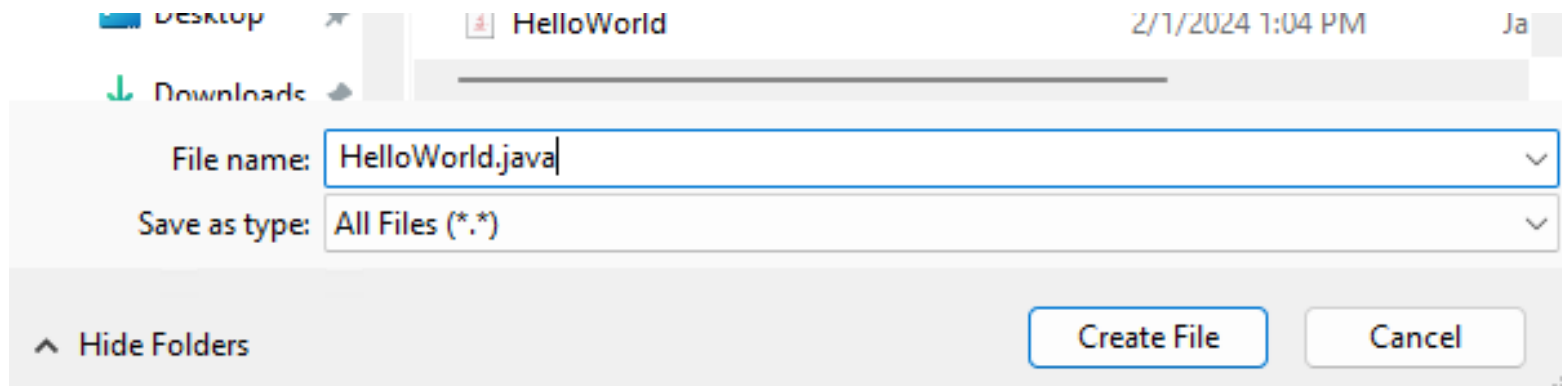
Code Editor

Step 2: Create New File (**java**) Build-in



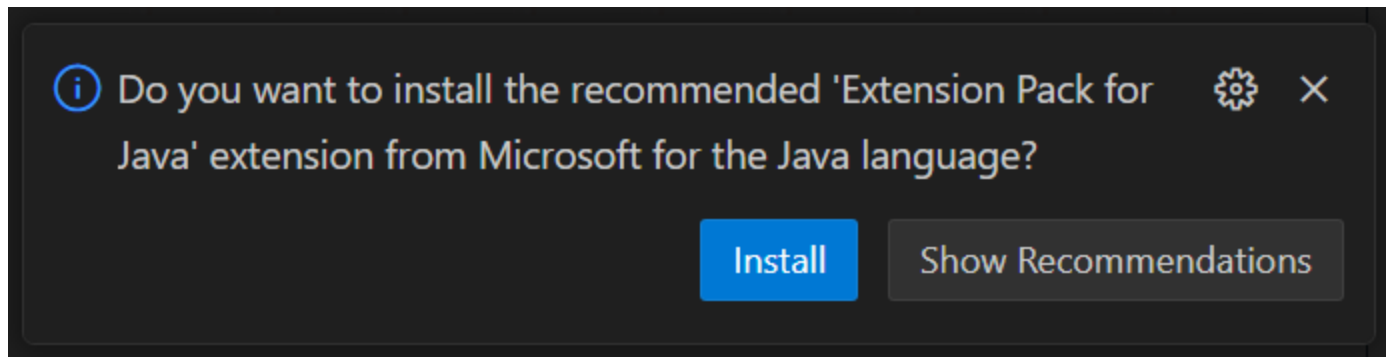
Code Editor

Step 4: **Name** of your class (**Capital** letter and **no space** in the name)



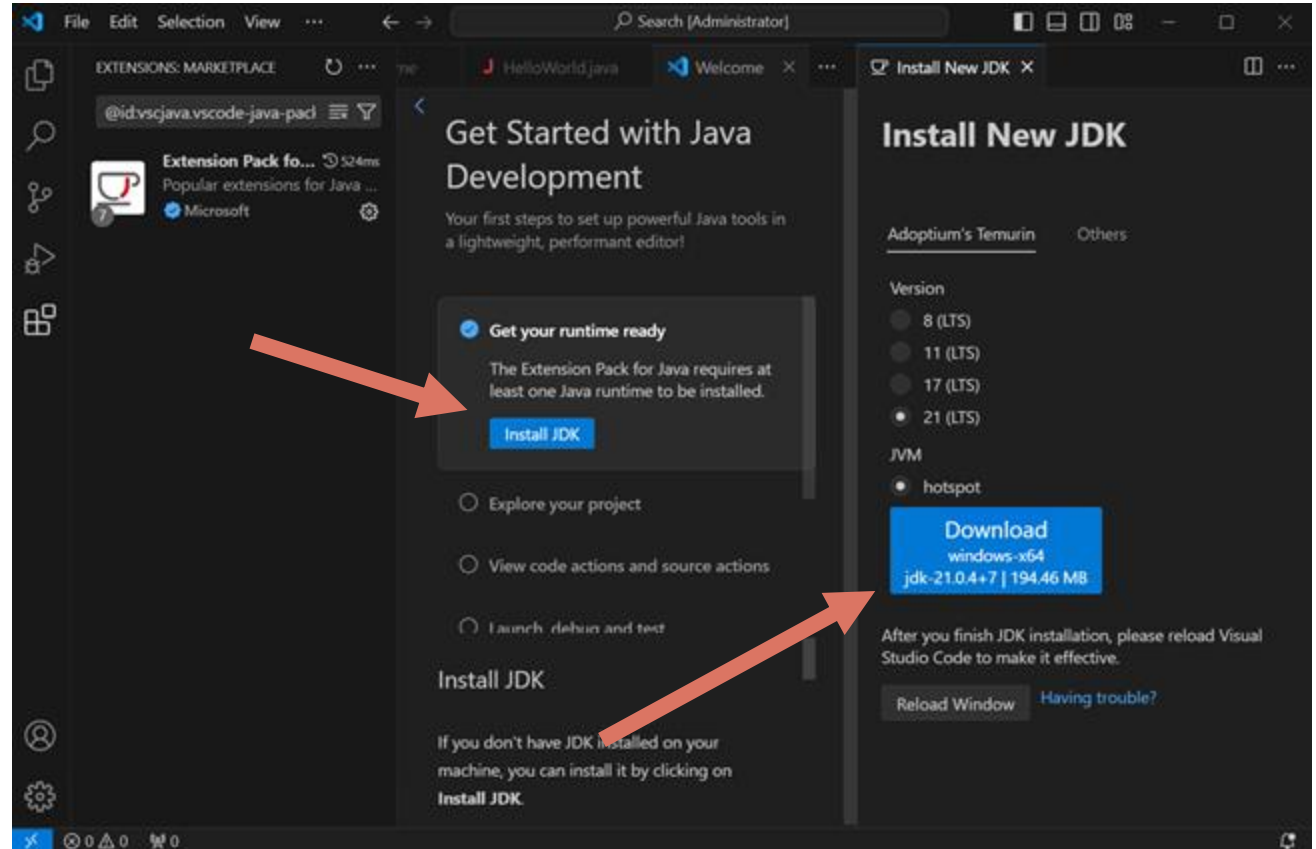
Code Editor

Step 5: When you create a .java file for the first time, the editor will offer to install the Extension pack for Java.



Code Editor

Step 6: When it finishes installing the extension, the editor will ask you to install the Java Development Kit (JDK)



```
public class HelloWorld {  
    public static void main(String[] args){  
        System.out.println("Hello World !");  
    }  
}
```

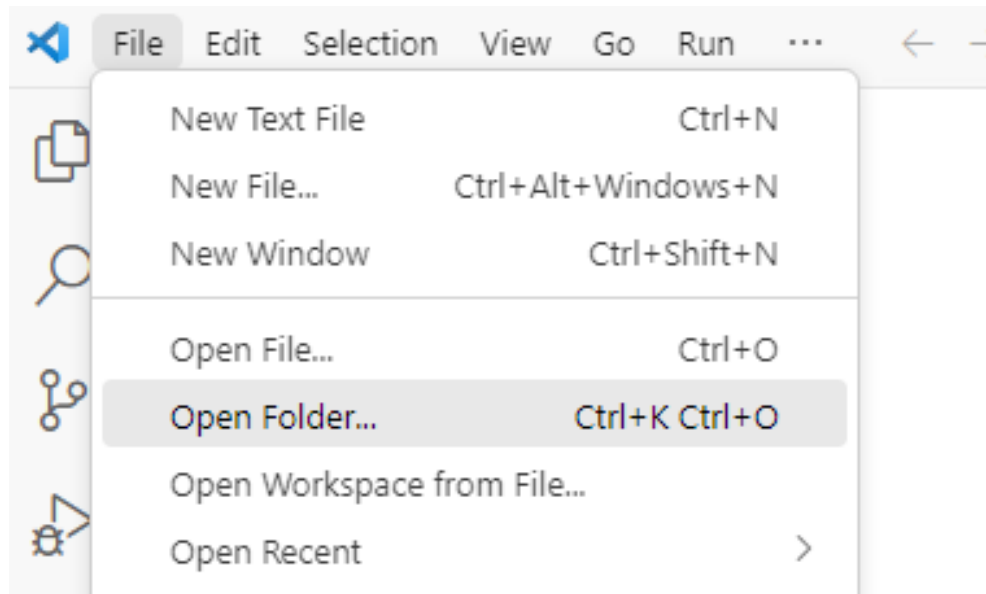
Code Editor

Open and run an existing program



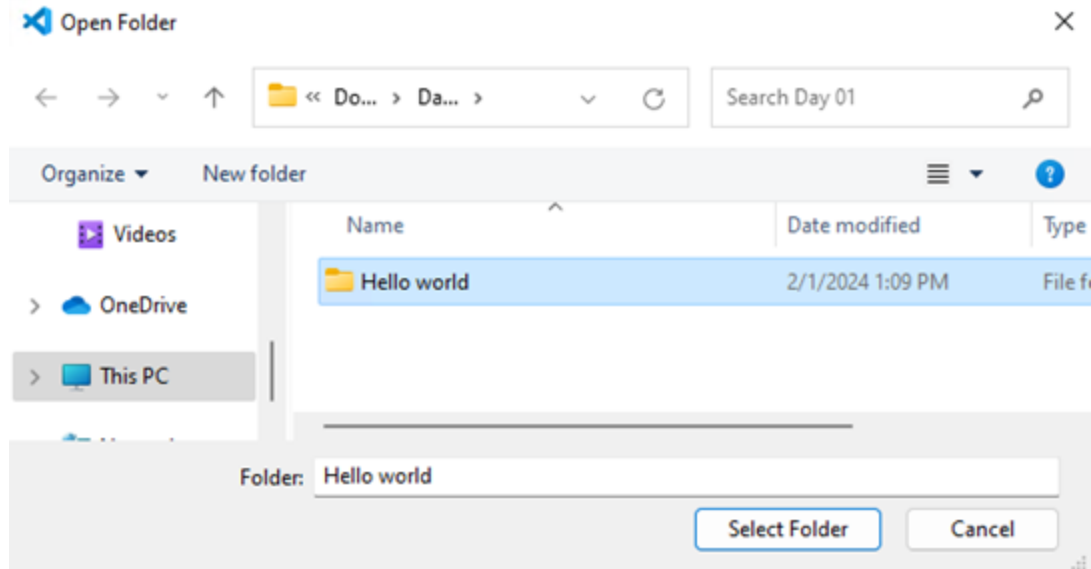
Code Editor

Step 1: **Open** the main folder.

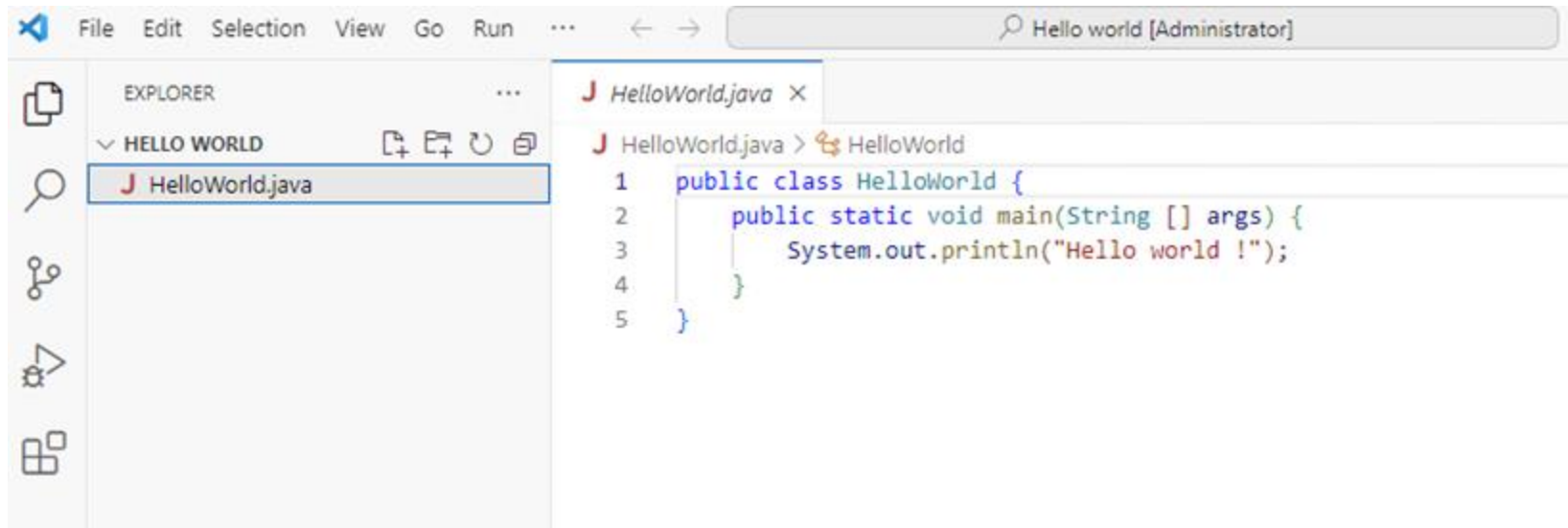


Code Editor

Step 2: **Choose** the main folder.

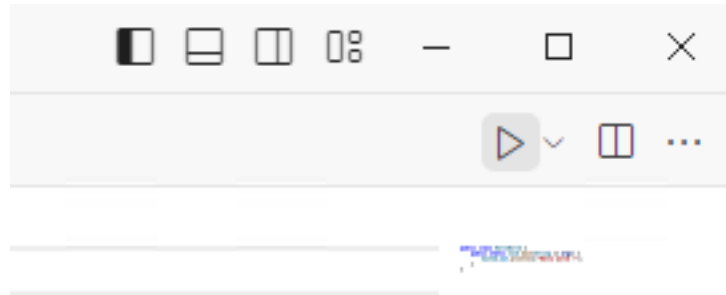


Code Editor



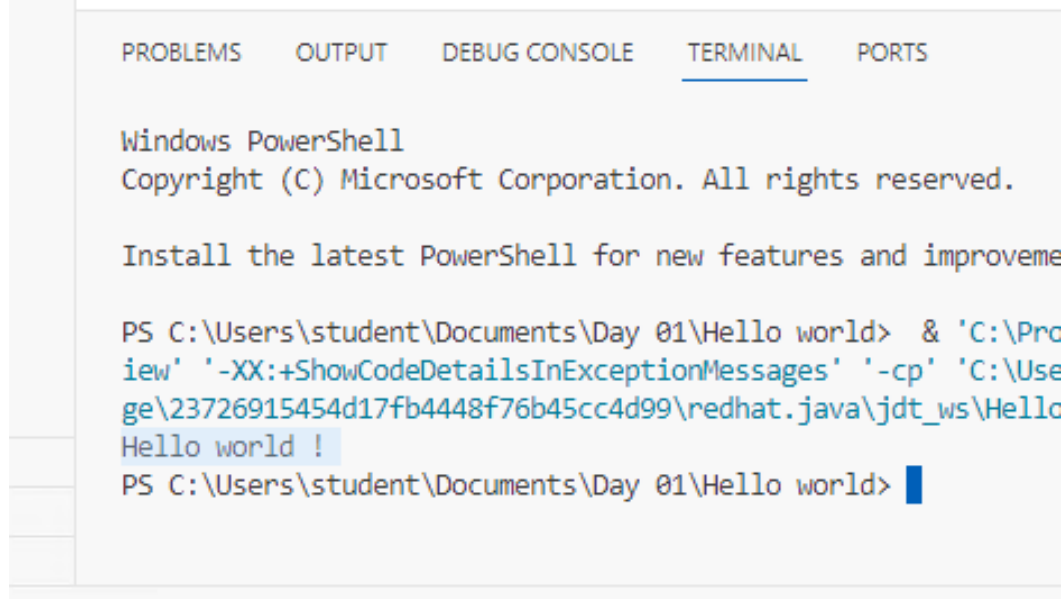
Code Editor

Step 3: **Run** the program



Code Editor

Step 4: See the **output**



The screenshot shows a terminal window within a code editor. The terminal has tabs at the top: PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The terminal text is as follows:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PowerShellLatest

PS C:\Users\student\Documents\Day 01\Hello world> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\student\Documents\Day 01\Hello world\redhat.java' jdt_ws\Hello world
Hello world !
PS C:\Users\student\Documents\Day 01\Hello world> 
```

Variables

A variable is a data container.

It can contain 1 information per time.

```
int age = 30;
```

Variables

```
data_type name = value ;
```

Variables

- Start with letter or `_`underscore
- Unique word in the code
- No special symbols
- No Java keyword

Variables

```
public class VariableNaming {  
    public static void main(String[] args) {  
        // Correct variable names following the naming rules:  
  
        // Rule: Starts with a letter.  
        // This variable name starts with the letter 'a'.  
        int age = 30;  
  
        // Rule: Starts with an underscore.(but not recommended)  
        // This variable name starts with an underscore.  
        double _salary = 50000.0;  
    }  
}
```


Variables

Use variable value

```
public class HelloWorld {  
    public static void main(String[] args) {  
        String helloWorld = "Hello World !";  
        System.out.println(helloWorld);  
    }  
}
```

Data Type

```
class DataTypes {
    public static void main(String [] args) {
        // integer types
        byte aSingleByte = 100; // -128 to 127
        short aSmallNumber = 20000; // -32,768 to 32,767
        int anInteger = 2147483647; // -2147483648 to 2147483647
        long aLargeNumber = 9223372036854775807L; // -92233720368547

        // decimal types
        double aDouble = 1.79769313; // 4.9E-324 to 1.797693
        float aFloat = 3.4028F; // 1.4E-45 to 3.40282

        // booleans
        boolean isWeekend = false;
        boolean isWorkday = true;

        // characters
        char copyrightSymbol = '\u00A9';

        System.out.println("This is the copyright symbol: " + copyrightSymbol);
        // This is the copyright symbol: ©

    }
}
```

Let's start with simple data types

```
public class HelloWorld {  
    public static void main(String[] args) {  
        String name = "Lisa";  
        char lastNameFirstLetter = 'H';  
  
        int year_of_birth = 1996;  
        double height = 1.57;  
  
        boolean hasPets = true;  
  
        System.out.println("Hello, my name is " + name);  
    }  
}
```

Operations

You can also do simple mathematical operations, like this:

```
int number1 = 10;  
int number2 = 15;  
int sum = number1 + number2;  
System.out.println(sum);
```

Operations

```
1 public class BasicOperations {
2     public static void main(String[] args) {
3         // Arithmetic Operations
4         int a = 10;
5         int b = 5;
6
7         int addition = a + b; // Addition
8         int subtraction = a - b; // Subtraction
9         int multiplication = a * b; // Multiplication
10        int division = a / b; // Division
11        int modulus = a % b; // Modulus (remainder of division)
12
13        System.out.println("Addition: " + addition);
14        System.out.println("Subtraction: " + subtraction);
15        System.out.println("Multiplication: " + multiplication);
16        System.out.println("Division: " + division);
17        System.out.println("Modulus: " + modulus);
18    }
19 }
```

Operations

```
1 public class BasicOperations {
2     public static void main(String[] args) {
3         // Arithmetic Operations
4         int a = 10;
5         int b = 5;
6
7         // Increment and Decrement
8         int increment = ++a; // Increments and then uses the variable
9         int decrement = --b; // Decrements and then uses the variable
10
11         System.out.println("Increment: " + increment);
12         System.out.println("Decrement: " + decrement);
13     }
14 }
```

Operations

```
1 public class BasicOperations {
2     public static void main(String[] args) {
3         // Arithmetic Operations
4         int a = 10;
5         int b = 5;
6
7         // Relational Operations
8         boolean isGreater = a > b; // Greater than
9         boolean isLess = a < b; // Less than
10        boolean isEqual = a == b; // Equal to
11
12        System.out.println("Is a greater than b? " + isGreater);
13        System.out.println("Is a less than b? " + isLess);
14        System.out.println("Is a equal to b? " + isEqual);
15    }
16 }
```

Operations

```
1 public class BasicOperations {
2     public static void main(String[] args) {
3         // Arithmetic Operations
4         int a = 10;
5         int b = 5;
6
7         // Logical Operations
8         boolean andOperation = (a > 5) && (b < 10); // Logical AND
9         boolean orOperation = (a > 10) || (b < 10); // Logical OR
10        boolean notOperation = !(a == b); // Logical NOT
11
12        System.out.println("Logical AND result: " + andOperation);
13        System.out.println("Logical OR result: " + orOperation);
14        System.out.println("Logical NOT result: " + notOperation);
15    }
16 }
```


Operations

Math library

- Predefined constants (PI)
- Functions (exponential, **power**, etc)
- Utility methods (round, random, etc)

```
public static void main(String[] args) {  
    double P = 1000; // Principal amount  
    double r = 0.05; // Annual interest rate (5%)  
    int n = 12; // Compounded monthly  
    int t = 5; // Time in years  
  
    double A = P * Math.pow(1 + (r / n), n * t);  
}
```

Now YOUR TURN !

Let's do exercises number 2

Input

Input is the process of providing data or information to a computer program during its execution.

Input

To obtain an input in java, you need a special “tool” called **Scanner** that can recognize the entered data.

Input

```
1 import java.util.Scanner;
2
3 public class BasicOperations {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6     }
7 }
```

Input

```
1 import java.util.Scanner;
2
3 public class BasicOperations {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         // getting data according to their type
8         String name = scanner.nextLine(); // String
9         int age = scanner.nextInt(); // int
10    }
11 }
```

Now YOUR TURN !

Let's do exercises number 3

Project Students - Step 1

Your program should store the following pieces of information for a student:

- First Name - String
- Last Name - String
- Birthday (Day of the Month) - int
- Birth Month - int
- Birth Year - int
- Course Registered - String

Project Students - Step 1

The program should print the requested information plus the student's age.

Output:

For now, we don't take month and day into account for the calculation.

To get the current year:

```
LocalDate today = LocalDate.now();  
int currentYear = today.getYear();
```

```
Enter first name: Ana  
Enter last name: Gaggero  
Enter birthday (day of month): 22  
Enter birth month: 10  
Enter birth year: 1982  
Enter course registered: Java
```

```
Student Name: Ana Gaggero  
Date of Birth: 22/10/1982  
Age: 42  
Course Registered: Java
```

Project Tic Tac Toe - Step 1

- Ask the names of both players
- Print the board before the first move

```
What is your name Player 1? Ana
What is your name Player 2? Juan

Ana will be X and Juan will be O

 1 | 2 | 3
 4 | 5 | 6
 7 | 8 | 9
```